

QUEUEING MODELS SIMULATION USING ARENA

United Dairy Farmers

Submitted By:

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1) Introduction

The United Dairy Farmers store remains open for 24 hours 7 days a week. It typically witnesses the rush around 9:00 PM to 11:00 PM every day. The UDF store offers dairy products to the customers which they can look around and put in their baskets to shop. The store had a self-service beverage counter for coffee, soft drinks and hot chocolate. The UDF store also offers Milkshakes to its customers, for which they have a dedicated Milkshake Maker staff member at the milkshake counter. Post selecting the dairy products, beverage or milkshake, the customers move to the billing counter to pay their bills where 2 cashiers serve them with payment.

2) Problem Statement

As per the staff at UDF, the UDF store typically witnesses the rush around 9:00 PM to 11:00 PM every day. Because of the rush, the average waiting time of customers in queues at the billing counter and milkshake counter increases considerably. Moreover, many of the customers leave the outlet and prefer to buy their dairy products, beverage and milkshakes from other stores nearby to avoid long queues at UDF, resulting in loss of potential revenue for the UDF store. This project aims to improve customer service operations at UDF by minimizing the customer waiting times and optimizing the utilizations of UDF staff members.

3) Objectives

The purpose of this project is to develop a simulation model of United Dairy Farmers store using Arena Software. The objective of building this project is to understand and optimize the following things: -

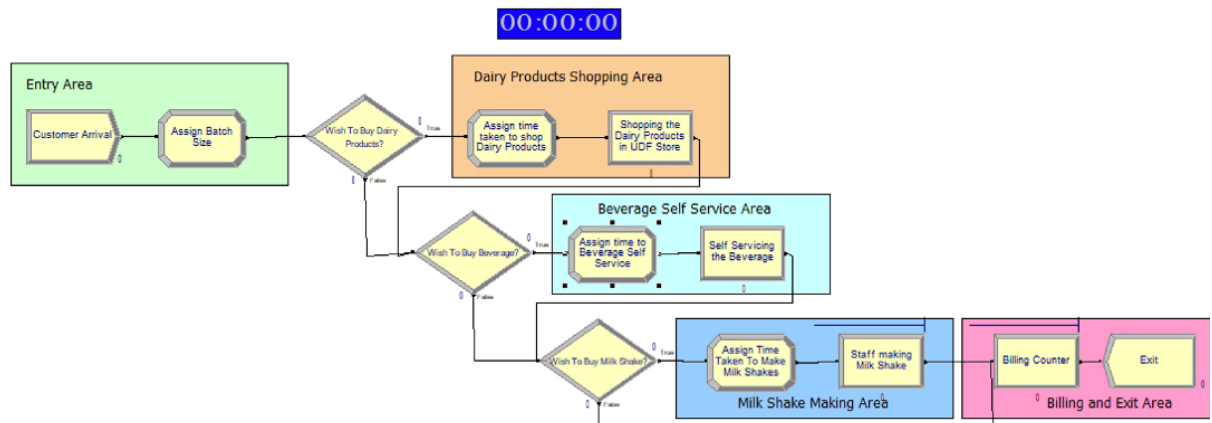
- Wait time in queue for the customers at the billing counter
- Wait time in queue for the customers at the MilkShake counter
- Utilization of UDF Staff performing different tasks
- Factors resulting in delay of customer service

4) Assumptions.

- a) Only 1 customer is served by 1 UDF staff at any given time.
- b) Customers come to UDF store individually or in a batch size of 2. If they arrive in batch of 2, they purchase similar items.
- c) Billing Counter Queue follows FIFO
- d) Milkshake Counter Queue follows FIFO

Key Features Analyzed: Customer Waiting Time, Percentage of Customers Served, Utilization of Resources, Queue Length.

5) Arena Diagram



- Customer inter arrival time: - A normal distribution followed
- Time taken to buy Grocery Items: Distribution followed: TRIA (4,16,24)
- Time taken at beverage self-service counter: A binomial distribution followed
- Time taken at milkshake counter: Distribution followed: TRIA (4,6,9)
- Billing Time: - A binomial distribution followed

6) Simulating the System

Run Setup

Run Speed

Run Control

Reports

Project Parameters

Replication Parameters

Array Sizes

Arena Visual Designer

Number of Replications:

50

Initialize Between Replications

☒ Statistics
 ☒ System

Start Date and Time:

☐ Saturday
 , May 10, 2025 6:12:12 PM

Warm-up Period:

0.0

Time Units:

Minutes

Replication Length:

2

Time Units:

Hours

Hours Per Day:

24

Base Time Units:

Minutes

Terminating Condition:

7) Results

Our model served 39 customers out of 46 customers who entered the UDF Store. On an average a customer stayed in store 21.9 minutes and the average waiting time was 19.53 minutes

6:17:03PM

Entities

May 10, 2025

Unnamed Project

Replications: 50

Replication 1

Start Time: 0.00

Stop Time: 120.00

Time Units: Minutes

Entity Detail Summary

Time

	NVA Time	Other Time	Total Time	Transfer Time	VA Time
Customer	0.00	0.00	21.89	0.00	2.36
Total	0.00	0.00	21.89	0.00	2.36

Other

	Number In	Number Out
Customer	46	39
Total	46	39

6:17:03PM

Entities

May 10, 2025

Unnamed Project

Replications: 50

Replication 1

Start Time:

0.00

Stop Time:

120.00

Time Units: Minutes

Customer

Time	Average	Half Width	Minimum	Maximum
Other Time	0.00	(Insufficient)	0.00	0.00
Wait Time	19.5337	(Insufficient)	0.00	58.0963
Total Time	21.8923	(Insufficient)	1.4866	61.1915
Transfer Time	0.00	(Insufficient)	0.00	0.00
NVA Time	0.00	(Insufficient)	0.00	0.00
VA Time	2.3586	(Insufficient)	1.2519	3.2856

Results based on queue show that for Billing Counter Queue average waiting time was 5.78 minutes.

6:21:23PM

Queues

May 10, 2025

Unnamed Project

Replications: 50

Replication 1

Start Time:

0.00

Stop Time:

120.00

Time Units: Minutes

Queue Detail Summary**Time**

	<u>Waiting Time</u>
Billing Counter.Queue	5.78
Staff making Milk Shake.Queue	0.42

Arena's resource detail summary reveals that the cashiers' utilization was 71% and Milkshake Maker staff's utilization was 18%.

6:38:56PM

Resources

May 10, 2025

Unnamed Project

Replications: 50

Replication 1

Start Time:

0.00

Stop Time:

120.00

Time Units: Minutes

Resource Detail Summary

Usage

	<u>Inst Util</u>	<u>Num Busy</u>	<u>Num Sched</u>	<u>Num Seized</u>	<u>Sched Util</u>
Cashier	0.71	1.41	2.00	72.00	0.71
MilkShake_Mak	0.18	0.18	1.00	3.00	0.18

8) Analysis

Utilization of Milkshake Maker staff (18%) is significantly less from the utilization of cashier (71%). Also, the average waiting time of customer is considerably high.

Our aim is to reduce the wait time of customers in current scenario without adding any additional financial burden for UDF store. One of the suggestions is to train Milkshake staff members with the skills required at billing counter so that during peak hours when there are long queues at billing counter, the Milkshake Maker staff member can temporarily perform duties of a cashier in between to reduce billing queue's wait time.

6:56:12PM

Entities

May 10, 202

Unnamed Project

Replications: 50

Replication 1

Start Time: 0.00

Stop Time: 120.00

Time Units: Minutes

Entity Detail Summary

Time

	NVA Time	Other Time	Total Time	Transfer Time	VA Time
Customer	0.00	0.00	20.36	0.00	2.40
Total	0.00	0.00	20.36	0.00	2.40

Other

	Number In	Number Out
Customer	48	35
Total	48	35

Unnamed Project

Replications: 50

Replication 1

Start Time:

0.00

Stop Time:

120.00

Time Units: Minutes

Customer

Time	Average	Half Width	Minimum	Maximum
NVA Time	0.00	(Insufficient)	0.00	0.00
Total Time	20.3624	(Insufficient)	1.4866	54.7500
Wait Time	17.9644	(Insufficient)	0.00	52.1710
Transfer Time	0.00	(Insufficient)	0.00	0.00
Other Time	0.00	(Insufficient)	0.00	0.00
VA Time	2.3979	(Insufficient)	1.2519	3.2856

Results show that the average waiting time of our updated model got improved from 19.53 minutes to 17.96 minutes, thus improvement by reduce of 1.59 minutes of waiting time. Also, the average time a customer stayed in store reduced from 21.9 minutes to 20.36 minutes.

Unnamed Project

Replications: 50

Replication 1

Start Time:

0.00

Stop Time:

120.00

Time Units: Minutes

Queue Detail Summary

Time

	Waiting Time
Billing Counter.Queue	2.36
Staff making Milk Shake.Queue	0.00

Results show that for Billing Counter Queue average waiting time improved from 5.78 minutes earlier to just 2.36 minutes now.

7:07:10PM

Resources

May 10, 2025

Unnamed Project

Replications: 50

Replication 1

Start Time:

0.00

Stop Time:

120.00

Time Units: Minutes

Resource Detail Summary**Usage**

	<u>Inst Util</u>	<u>Num Busy</u>	<u>Num Sched</u>	<u>Num Seized</u>	<u>Sched Util</u>
Cashier	0.77	1.53	2.00	78.00	0.77
MilkShake_Mak	0.82	0.82	1.00	40.00	0.82

Result reveals that the Milkshake Maker staff's utilization jumped from 18% in previous model to 82% in updated model.

9) Conclusion

We collected 2-hour window UDF data during peak interval and based on certain assumptions we developed a simulation model in Arena software. We simulated and analyzed the Arena results of our model. To achieve our goal of reducing the wait time of customers in the current scenario, we made a few changes in our original model for improvements. We observed that by training the Milkshake Maker staff member at UDF store, we can reduce the average wait time by 1.59 minutes. With the improved model we can also improve the resource utilization of the store and reduce inconvenience being caused to customers during peak hours.